JCLAP · CLAP INSTRUMENT

JSynth

The flocking synth. User Manual



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0.10.0	CLAP · Instrument · MPE	macOS · Windows · Linux	October 2026

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About this manual

Chapters 2 and 3 get you installed and playing in a few minutes. Chapter 4 tours the panel, and Chapter 5 explains how a note becomes a sound, which is the quickest way to understand what every control does. Chapters 6 to 13 cover the sky, the parts and every panel. Chapter 14 lists the factory presets and Chapter 15 has recipes to start from. The appendix lists every parameter with its exact range and default, plus specifications and troubleshooting.

Throughout the manual, control names are set in **bold** and panel labels as they appear on screen, such as **TUNE PULL**. Every control is a host parameter, and every one that moves continuously can be modulated by the host. A tag beside a parameter means:

ROUTE A route can move it, per note, and the host can modulate it per note too.

CHAPTER 01

Welcome to JSynth

A two-part MPE synth where every note is a creature, and the creatures flock.

Play a note on JSynth and a creature hatches in the sky in the middle of the panel. Your note is its **lure**: low notes pull their creatures to the left, high notes to the right, and sliding your finger up an MPE controller lifts it higher. Press harder and the creature flies faster and burns brighter. Let go and it flies off as its sound fades, trailing ink.

While they are held, the creatures do what flocks do. They gather, turn together, keep a little apart, and wander. And because each creature is a voice, how it flies can become how it sounds: how fast it goes, how near its nearest neighbour is, which way it heads and how hard it turns are all modulation sources, as exact as an envelope. Creatures can also hear each other directly. A neighbour's oscillator can frequency-modulate this one, creatures flying the same way drift toward pure intervals, and a hunter that catches its prey rings it.

Underneath is a serious synth. Each of the two parts, **A** and **B**, is its own species with its own patch: three oscillators with analog and FM waves, a 3×3 FM grid, noise, drive, two filters, three envelopes, four LFOs and sixty-four routes. Nothing the flock does reaches the sound unless you route it, so a plain patch is plain.

At a glance

- **Two parts**, layered, split, by velocity, by MPE zone, or crossfaded by expression. Up to 16 voices each.
- **Three oscillators** per voice: sine, triangle, saw, square, pulse, the TX81Z's FM waves and noise, pitched by note, ratio or fixed Hz.
- **A 3×3 FM grid**, phase or exponential, whose diagonal is each oscillator's feedback.
- **Two filters** per voice: SVF up to 48 dB, notch, peak, ladder and comb, in serial or parallel, with each oscillator picking its filter.
- **MPE** and CLAP note expressions: velocity, press, glide, slide and release velocity, each with its own curve and smoothing.
- **The flock**: cohesion, alignment, separation, speed, wander, lure and range, with four behaviours: Shoal, Scatter, Hunt and Flee.
- **Four flock sources** per creature: Speed, Nearest, Heading and Turn.
- **Between creatures**: Neighbour FM, Tune Pull and Bite.
- **MAP**: touch a source, drag a knob. No matrix page, and up to 64 routes per part.
- **92 factory presets**, from basses, keys and pads to patches that move on their own and use both parts.

THE FLOCK IS THE ENGINE'S

The flock flies on the audio thread, at a fixed rate, inside the synth. What you see in the sky is exactly what the voices hear, and a render plays back the same every time from the same start.

CHAPTER 02

Installation

JSynth is a CLAP plug-in. It ships in the JClap bundle alongside the other JClap plug-ins.

Requirements

Plug-in format	CLAP 1.2 (instrument, stereo). Your host must support CLAP, for example Bitwig Studio, REAPER, FL Studio or Studio One.
macOS	Universal binary: macOS 10.15 or later on Intel, 11 or later on Apple Silicon.
Windows	64-bit Windows.
Linux	64-bit, X11.
Controller	Any MIDI keyboard. An MPE controller (a Seaboard, Linnstrument, Osmose or similar) or a host that sends CLAP note expressions brings out the most.

Installing

01 **Download**

Grab the zip for your operating system from the latest release at `github.com/jeffmhopkins/Jclap/releases/latest`. It contains every JClap plug-in.

02 **Copy the plug-in**

Copy `JSynth.clap` (a bundle on macOS) into your CLAP folder:

macOS	`~/Library/Audio/Plug-Ins/CLAP`
Windows	`C:\Program Files\Common Files\CLAP`
Linux	`~/.clap`

03 **Rescan**

Restart your host or rescan plug-ins. In Bitwig Studio this is under *Settings → Locations*. JSynth appears as an instrument.

MACOS: BROWSER DOWNLOADS ARE QUARANTINED

The plug-ins are ad-hoc signed but not notarized, so macOS quarantines files downloaded through a web browser and some hosts (Bitwig among them) silently skip them. The simplest install is from Terminal, since files fetched with `curl` are not flagged:

```
curl -sL https://github.com/jeffmhopkins/Jclap/releases/latest/download/jclap-macOS.zip -o /tmp/jclap.zip
&& mkdir -p ~/Library/Audio/Plug-Ins/CLAP && ditto -x -k /tmp/jclap.zip ~/Library/Audio/Plug-Ins/CLAP
```

If you already downloaded with a browser, clear the flag once after copying:

```
xattr -dr com.apple.quarantine ~/Library/Audio/Plug-Ins/CLAP/*.clap
```

Outputs

JSynth has one note input and three stereo outputs: **Main**, **Part A** and **Part B**. Normally both parts play on Main. Set a part's **Output** to *Own* (a host parameter) and it plays on its own port instead, so you can process the two species separately in your host.

CHAPTER 03

Quick Start

Five minutes from an empty instrument slot to a flock you can hear.

01 **Load JSynth and play**

Put JSynth on an instrument track and play a few notes. Each note hatches a creature at its lure: low notes on the left, high on the right. Hold a chord and watch them gather; let go and they fly off as they fade. No keyboard to hand? Click in the sky: left to right picks the note (C2 to C6), and dragging bends it.

02 **Pick a preset**

Click the preset name in the header to open the menu, grouped by instrument family from *Bass*, *Leads* and *Keys* through *Pads* and *Drums and FX*, then the showcases: *Motion*, *Layers and Splits* and *Flock*. Or click the arrows, or scroll over the bar, to step through them. Try *Tine Piano*, then *Shifting Glass*, then *Breath to Brass* with some pressure, then *Predator and Prey*.

03 **Let the creatures hear each other**

Load *Init* and turn **OSC 2**'s **LEVEL** up to about half, since Neighbour FM plays through Osc 2. Then open the **FM** tab at the top left and turn up **FLOCK FM**. Now hold a chord: when creatures fly close together they frequency-modulate each other, and the sound rings and shifts as the flock moves. Raise **TUNE PULL** too and creatures flying the same way drift into just intonation.

04 **Map a flock source**

Click **SPEED** under the sky, which turns **MAP** on with Speed armed. Then drag Filter 1's **CUTOFF** up on the **FILTER** panel at the left. A ring in Speed's colour shows the route while Speed is armed. Press **MAP** again to finish. Now the faster a creature flies, the brighter it sounds; play harder (Press makes creatures faster) to hear it.

05 **Add a second species**

Set the play mode in the header to **LAYER** and click **PART B**. Part B is a whole second patch, and by default it hunts: its creatures chase part A's. On part B's **FM** tab, raise **BITE**, and every creature B catches rings with its hunter's tone.

06 **Save it**

Click **SAVE** in the preset bar, type a name and press Enter. Your preset appears under *User* in the preset menu. Settings are also saved with your project automatically.

> **TIP**
> Every control has a tooltip. Hover over anything on the panel for a one-line reminder of what it does.

CHAPTER 04

The Panel

The sound sits down the left in three big panels that draw what they do, the sky fills the right, and the voice and the output sit under it. Everything edits the part you have chosen.



1. **Preset bar.** Step with the arrows, click the name for the menu, **SAVE** your own.
2. **PART A / PART B.** Which species the panels edit.
3. **Play mode.** Which part each note plays.
4. **MAP.** Mapping: touch a source, drag a knob. Touching a source turns it on too.
5. **Touch chips.** VELOCITY, PRESS, GLIDE, SLIDE, RELEASE: sources that light as you play.
6. **OSCILLATORS / FM.** The three oscillators with their waves drawn; the FM grid, the links between creatures and the noise.
7. **FILTER.** Both filters' response, drawn live, and their controls.
8. **ENVELOPES / LFOS.** ENV 1 AMP, ENV 2 and ENV 3; LFO 1 and 2.
9. **The sky.** Both species, their ink, and the links between them. Click to play.
10. **Flock sources.** The creatures' own modulation sources, and Random.
11. **RINGS, ROUTES.** Lure rings and the route list.
12. **VOICE / FLOCK / MOTION / TOUCH.** The voice, how the species flies, the wingbeat and the wind, and how each touch responds.

13 **Meters.** The levels of A, B and the output as they play, with each one's setting marked.

14 **SYNTH / SKY.** The two views: the panels beside the sky, or the sky alone.

Working with the controls

Every knob, switch and menu is a host parameter, so everything can be automated. Knobs work the same everywhere on the panel:

ACTION	ON A KNOB
Drag up / down	Change the value. A popup shows the value while you drag.
Shift **+ drag**	Fine adjustment.
Scroll wheel	Step the value up or down.
Double-click	Reset to the default value.
Right-click	Menu with *Reset to default* and *Enter value…* to type an exact number.

In MAP mode, with a source armed, each knob that source moves draws a **route ring** in its colour outside its own arc, spanning how far the source moves it. Outside MAP the knobs show no rings; the route list shows every route. In MAP mode, knobs behave differently: see *Chapter 13*.

Colours

Part A is drawn in ● orange and part B in ● blue, in the sky and on the panels. Every modulation source has its own colour, the same on its chip, its route rings and in the route list, and MAP and the route list are ● pink.

Panel size

The panel is 1648 × 940 at 100%. Right-click any empty part of the sky and choose *UI Size* to scale it to 75%, 100%, 125%, 150% or 200%; at 75% it fits a 1440 × 900 laptop screen. The size you pick is remembered the next time the editor opens.

CHAPTER 05

How JSynth Works

A note becomes a voice and a creature at once. The voice makes the sound; the creature flies, and how it flies can shape the sound.

One voice

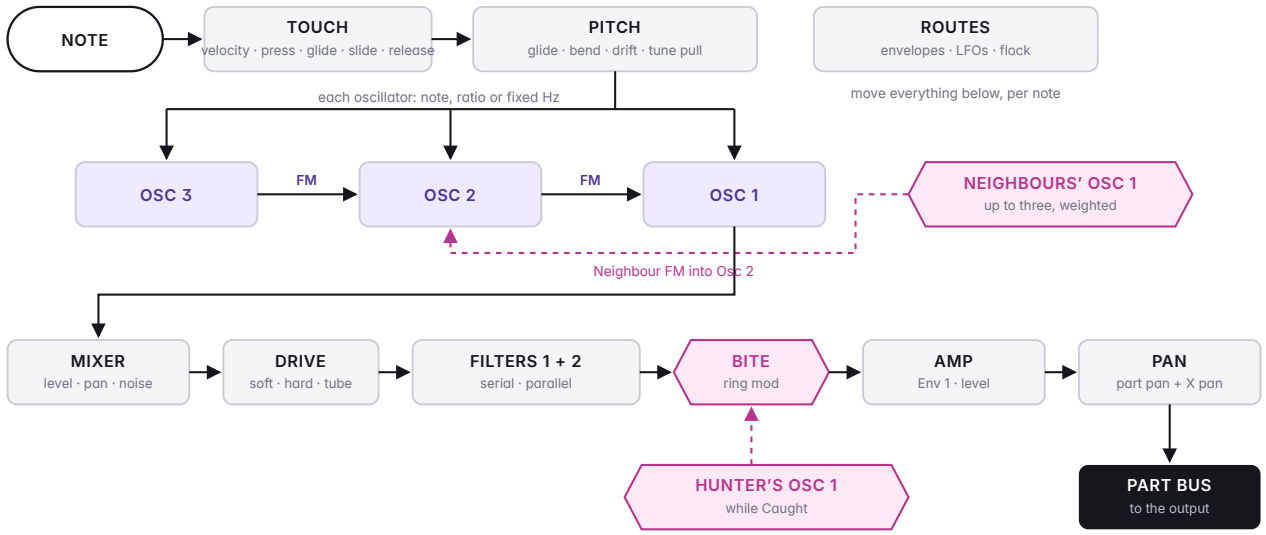


Fig. 5.1 One voice, at the oversampled rate. The pink paths are the couplings to other creatures: a neighbour's Osc 1 into Osc 2, and a hunter's Osc 1 ringing the voice.

Each voice runs three **oscillators**, in the order 3, 2, 1, so Osc 3 can modulate Osc 2, and both can modulate Osc 1, within the same sample. The **FM grid** sets how much each one modulates each other one. The **mixer** sets each oscillator's level and pan and adds noise. Then comes **drive**, then two **filters**, then the **amp**, whose shape is Env 1, and finally the voice's place in the stereo field. Every voice of a part sums to that part's bus, and the two parts sum to the output.

The control rate

Envelopes 2 and 3, the LFOs, the touch, the routes and the flock all work at a control rate of one tick every 32 samples (1,500 a second at 48 kHz). Every value the audio reads ramps smoothly across each tick, so nothing zippers, and knobs are smoothed over 10 ms. The amp envelope runs every sample. Because the ticks are counted across the host's blocks, JSynth sounds the same at any buffer size.

The creature

As a voice starts, its creature hatches at the note's lure. The flock steps 500 times a second. Each step, every creature feels its neighbours (cohesion, alignment, separation), its lure, its own wander, the wind and its wingbeat, and moves. Then the flock works out each creature's four sources and who can hear whom. Each voice reads its creature's sources smoothly between steps.

Routes

A **route** connects a source to a destination with an amount. Each part has 64 of them. Each tick, every voice works out its routes from its own sources, so the same route can push one note's filter open and leave another's closed: an envelope, a finger's pressure or a creature's speed is always that note's own. A route moves a parameter within its own range, so +100% from a source at full takes any knob from its lowest to its highest.

Quality

The voices run at 2× the host's rate by default, which keeps FM and hard waves clean, and are brought back down by a steep, latency-free filter. The **Quality** parameter (a host setting) chooses 1×, 2× or 4×. 1× saves CPU; 4× is for bright FM at low sample rates.

> **WHERE TO READ MORE**
> `plugins/jsynth/ARCHITECTURE.md` in the source repository walks through every stage with the numbers the code uses.

CHAPTER 06

The Sky

Both species fly here, all the time. The sky is the synth's own flock, drawn as it is.

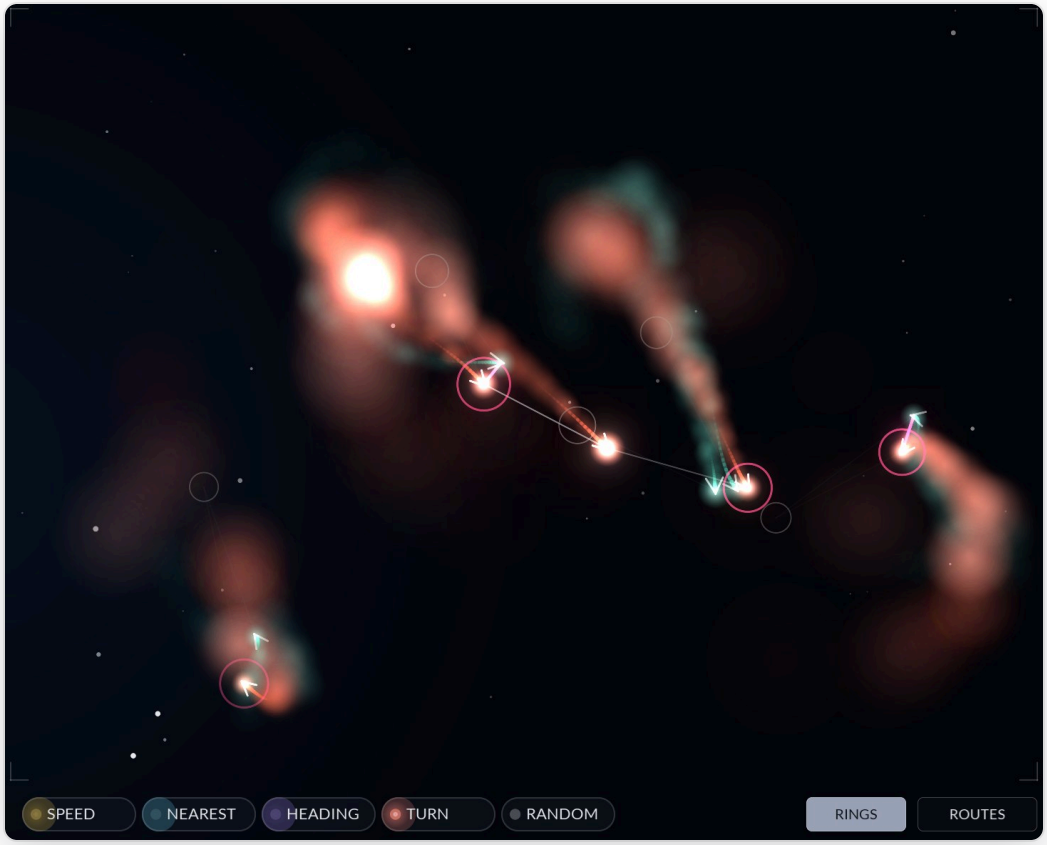


Fig. 6.1 Part A's creatures (orange) and part B's (teal) in a layered patch, with their ink. Faint grey rings are lures; a pink ring marks a creature a hunter has caught; thin lines join neighbours that hear each other.

What you see

Creatures A glowing body and a pair of wings that beat with its Wingbeat LFO. It is brighter the harder you press, and fades with its sound after you let go. The part you are editing flies in front; the other is dimmed.

Ink Every creature sheds ink behind it, which blooms and fades by the part's **TAIL** and **SPREAD**. Creatures flying through ink stir it.

Lure rings Where each held note pulls its creature, with a faint line to it. **RINGS** under the sky shows or hides them.

Links A thin line from a creature to the neighbour it hears most, brighter the more **FLOCK FM** it gets, and a line from a hunter to what it has caught.

Caught A pink ring around a creature a hunter holds.

Playing the sky

Click anywhere in the sky to play a note: left to right runs from C2 to C6, and how high you click sets Slide. Drag while holding to bend the note and change Slide; let go to release it. The notes play on whichever parts the play mode sends them to, just like notes from your keyboard.

The flock sources

The row of chips under the sky holds each creature's four sources, and **RANDOM**, a new value for each note. Each chip's light shows the newest creature of the part you are editing. They are always each note's own when used in a route.

SOURCE	WHAT IT MEASURES
SPEED	How fast the creature flies, against the fastest it could at full Press (0 to 100%).
NEAREST	How near its nearest neighbour is: 100% is touching, 0 is at the edge of **RANGE** or beyond.
HEADING	How much it flies upward: 100% straight up, 50% level, 0 straight down. Smooth all the way round, so it never jumps.
TURN	Which way and how hard it is turning: up to +100% for a hard right turn on screen, down to −100% for a hard left, 0 flying straight. Smoothed over about 170 ms, so the wander's jitter and the wingbeat's bob barely reach it.

Two views

The switch at the top right picks how much of the panel you see. **SYNTH**, as this manual shows it, puts the sound down the left beside the sky. **SKY** folds everything away but the switch so the sky fills the panel, which is lovely on a second screen while you play. The view stays as you left it when the editor is closed and opened again.

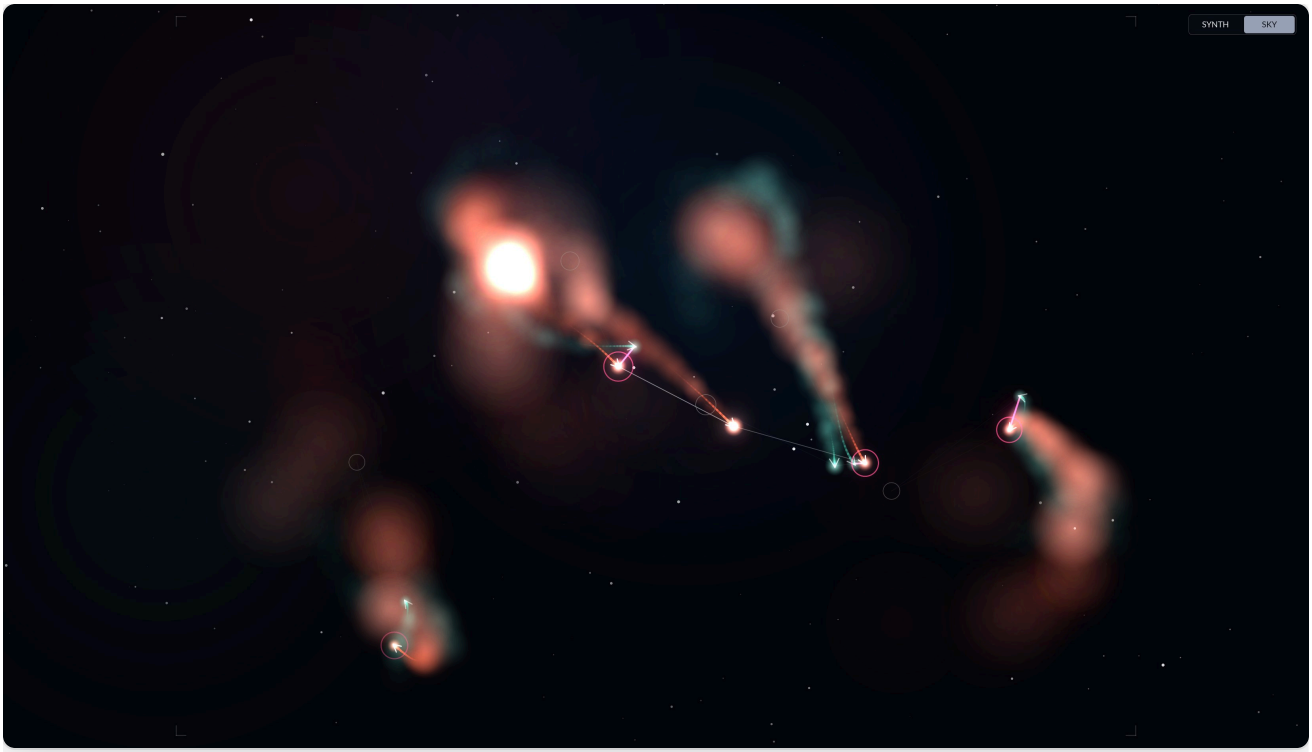


Fig. 6.2 SKY: the flocks alone.

CHAPTER 07

Parts and Playing

Two species, one keyboard. The header decides which notes go where; the VOICE tab under the sky sets how each part plays them.



Fig. 7.1 The header: logo, preset bar, PART A / PART B, play mode, MAP, the touch chips and the view switch.

Two parts

Parts A and B are complete, separate patches with the same controls. **PART A** and **PART B** in the header choose which one the panels show and edit; the sky always shows both. A new instance starts with A shoaling and B hunting it an octave down, so layering them gives you a predator from the start.

Play modes

MODE	WHICH PART PLAYS A NOTE
SINGLE	Part A only. The default.
LAYER	Both, every note.
SPLIT	B below the **Split Key** (C4 by default), A from it up.
MPE ZONES	MPE member channels 2 to 8 play A and 9 to 16 play B. Channel 1 plays both.
VELOCITY	B for notes struck at or above the **Velocity Split** (80), A for softer ones.
EXPRESSION	Both, crossfaded per note by Slide or Press (the **Crossfade Source**): at 0 you hear A, at full you hear B.

Split Key, Velocity Split and Crossfade Source, and each part's Key Low and Key High range, are host parameters. A note outside a part's key range never plays that part.

VOICE



The VOICE tab draws what its knobs set: the stereo field (the part's pan, its X Pan spread, the output's width and where its creatures sit), the voices in use, and at the right the **meters**. These show the levels of A, B and the whole output as they play, from −60 to +6 dB, glowing brighter the louder they are and red above 0 dB. A white tick on each marks where its LEVEL or MASTER is set. The part you are editing is lit; the other is dimmed.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
POLY / MONO / LEGATO	`Voice mode`	`Poly`	Mono plays one note at a time and restarts it; Legato glides to a new note without restarting while one is held. Both go back to the last note still held when you let go.
Voices (– 8 +)	`1 – 16`	`8`	Polyphony: how many notes the part sounds at most. The oldest released note is taken first, then the oldest held, with a 3 ms fade.
PAN	`±100%`	`0`	The part's pan. ROUTE
X PAN	`0 – 100%`	`0%`	How far each creature's place in the sky, left to right, pans its voice. At 100% a creature at the left edge plays hard left. ROUTE
WIDTH	`0 – 200%`	`100%`	The whole output's stereo width: 0% is mono, past 100% wider. Shared by both parts.
GLIDE	`0 ms – 20 s`	`0`	Portamento time. ROUTE
BEND	`1 – 96 st`	`48 st`	The MPE bend range: how far a finger slides on a member channel.
TRANSP	`±48 st`	`0 (B: -12)`	Transpose, in semitones.
VELOCITY	`0 – 100%`	`100%`	How much velocity sets the level. At 0 every note plays at full level.
LEVEL	`-60 – +6 dB`	`0 dB`	The part's level. ROUTE

CONTROL	RANGE	DEFAULT	WHAT IT DOES
MASTER	`-60 – +6 dB`	`-6 dB`	The whole output's level. Shared by both parts.

Host parameters that complete the part: **Fine** (±100 cents), **Glide Mode** (Always, or Legato: glide only from a note still held), **Key Low** and **Key High**, and **Output** (Main or Own).

MPE and MIDI

JSynth takes CLAP notes, plain MIDI and MIDI MPE.

- **Channel 1** is the MPE manager channel and a plain keyboard's channel. Its pitch bend moves every note by the **MGR BEND** range (2 semitones by default, on the TOUCH tab). CC 1 is the Mod Wheel source and CC 64 is sustain. If it sends channel pressure or CC 74, those reach every note too.
- **Channels 2 to 16** are member channels, each carrying one note's own bend (by the part's **BEND** range), pressure and CC 74.
- **CLAP notes** use note expressions instead: tuning, pressure and brightness.

Hosts that support CLAP per-note modulation can also modulate any parameter tagged ROUTE on a single note.

CHAPTER 08

Oscillators and FM

Three oscillators per voice, from soft sines to band-limited saws, and a grid that lets each one modulate the others.



Fig. 8.1 The OSCILLATORS panel: a row for each oscillator, with its wave drawn as its Wave and Shape make it.

An oscillator

CONTROL	RANGE	DEFAULT	WHAT IT DOES
NOTE / RATIO / HZ	`Pitch mode`	`Note (Osc 2, 3: Ratio)`	**Note** follows the keyboard with an offset; **Ratio** runs at a multiple of the note, the classic FM way; **Hz** holds a fixed frequency whatever you play.
Wave	`15 waves`	`Saw (Osc 2, 3: Sine)`	See the table below.
RAND / SYNC	`Retrigger`	`Sync`	**SYNC** starts each note's wave at its **Phase**; **RAND** at a random place, so no two notes start alike.
PITCH	`±48 st`	`0`	Note mode: the offset in semitones. ROUTE
RATIO	`×0.25 – ×16`	`×1 (Osc 3: ×3)`	Ratio mode (the knob reads RATIO): the multiple of the note. Routes move it by musical intervals. ROUTE
FREQ	`1 Hz – 10 kHz`	`440 Hz`	Hz mode (the knob reads FREQ): the fixed frequency. ROUTE
FINE	`±100 ct`	`0`	Fine tune, in every mode. ROUTE
SHAPE	`0 – 100%`	`50%`	Pulse width for Square and Pulse; for the other waves (except Saw), a phase distortion that squeezes or stretches the first half of the cycle. 50% is the plain wave. ROUTE

CONTROL	RANGE	DEFAULT	WHAT IT DOES
LEVEL	`0 – 100%`	`100% (Osc 2, 3: 0%)`	How loud it is in the mix. An oscillator at 0 can still modulate others. ROUTE
PAN	`±100%`	`0`	Its place in the voice's stereo image. ROUTE
FILTER	`1+2, F1, F2`	`1+2`	Which filter it goes into. See *Routing* in Chapter 09.

Host parameters per oscillator: **Sync** (Osc 2 and 3 restart with Osc 1), **Phase** ROUTE, **Bend** (how much the pitch bend moves it, 100% by default) and **Key Track** (Off pitches it from middle C whatever you play).

The waves

WAVE	CHARACTER
Sine	Pure. The classic FM operator.
Triangle	Soft, odd harmonics.
Saw	Bright and full. Band-limited, so it stays clean high up and under FM.
Square, Pulse	Hollow to nasal. Shape sets the width: 10 to 90% for Square, 5 to 50% for Pulse.
Half Sine … Alt Camel	The Yamaha TX81Z's seven other waves: Half Sine, Abs Sine, Quarter, Alt Sine, Alt Half, Camel and Alt Camel. Sines with parts of the cycle cut, folded or doubled, which give FM a whole family of brighter, buzzier tones.
White, Pink	Noise, as an oscillator: it can be filtered, panned and even used as an FM source.
Random	A different one of Sine to Alt Camel for each note, so a run of notes changes colour as it goes.

OSCILLATORS

FM



The **FM** tab draws what comes out of Osc 1, the oscillator you hear on its own, for a C4, with a chip for each FM cell that is turned up. The grid reads **row into column**: the knob in row 2, column 1 (**2 > 1**) is how much Osc 2 modulates Osc 1. The diagonal, **FB 1** to **FB 3**, is each oscillator's feedback on itself. Each knob is squared inside, so the first half of its travel is gentle.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
FB, 1 > 2 … 3 > 2	0 – 100%	0%	The nine cells. ROUTE
PHASE / EXP	FM type	Phase	**Phase** is DX-style phase modulation, bright and stable in pitch. **Exp** moves the pitch itself, up to four octaves either way: growly, and it bends.
FLOCK FM	0 – 100%	0%	Neighbours' Osc 1 into this creature's Osc 2. ROUTE
TUNE PULL	0 – 100%	0%	Creatures flying the same way drift toward pure intervals. ROUTE
BITE	0 – 100%	0%	Ring mod while this part's hunters hold their prey. ROUTE
NOISE	0 – 100%	0%	The mixer's noise level. ROUTE
COLOR	±100%	0	Pink (−100%) to white (+100%). ROUTE
DRIFT	0 – 100%	0%	Each creature's own slow wander in pitch, up to ±15 cents. ROUTE

Order and feedback

Within each sample the oscillators run 3, then 2, then 1. So the downward cells (3 > 2, 3 > 1 and 2 > 1) hear this sample's modulator, and the upward ones (1 > 2, 1 > 3, 2 > 3) hear the last sample's. In practice this means you can build any of the classic stacks: 3 > 2 > 1 as a three-operator stack heard on Osc 1, or 2 > 1 and 3 > 1 as two modulators on one carrier. Feedback averages its last two samples, as the DX7 did, so it brightens toward a saw and then a rasp rather than straight into noise.

BETWEEN CREATURES (Flock FM, Tune Pull and Bite) is explained in *Chapter 11*.

CHAPTER 09

Filters and Drive

Two filters per voice, from a clean 48 dB low pass to a ringing comb, after a drive stage.

LEFT

FILTER



The graph draws both filters' response together, Filter 1 in orange and Filter 2 in purple. Drag a numbered handle across to set its cutoff and up or down for its resonance; the wheel sets its drive and a double-click resets it. In MAP mode, with a source armed, the drag sets that source's route to the cutoff instead.

While a note plays, the curves follow what the newest note's filters are really doing: its envelopes, LFOs, routes and key tracking all move them, and a faint line stays where the knobs alone would put them. The handles stay on the knobs' settings.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
SERIAL / PARALLEL	`Routing`	`Serial`	See below.
NOISE INTO	`1+2, F1, F2`	`1+2`	Which filter the noise goes into.
Type	`8 types`	`Low Pass (F2: Ladder)`	See the table below.
Slope	`12 – 48 dB`	`24 dB`	Low, high and band pass only.
CUTOFF	`20 Hz – 20 kHz`	`1.26 kHz (F2: 20 kHz)`	For a comb, its tuning. ROUTE
RES	`0 – 100%`	`20% (F2: 0%)`	Resonance; for a peak, its boost; for a comb, its feedback. ROUTE

CONTROL	RANGE	DEFAULT	WHAT IT DOES
DRIVE	`0 – 100%`	`0%`	Saturates the filter's own input. ROUTE
KEY	`0 – 100%`	`0%`	Key tracking: at 100% the cutoff moves an octave per octave, around middle C. ROUTE
Drive type	`Soft, Hard, Tube`	`Soft`	The drive before both filters.
PRE DRIVE	`0 – 100%`	`0%`	Up to 24 dB into the shape, with the level brought back down. ROUTE
BALANCE	`±100%`	`0`	In Parallel, turns Filter 1 or Filter 2 down. ROUTE

Filter types

TYPE	SOUND
Low Pass	The workhorse: a state variable filter, cascaded for 12, 24, 36 or 48 dB per octave, with a resonance peak up to a whistle.
High Pass	The same, from the other side: thins and brightens.
Band Pass	Only a band around the cutoff. Narrower and more vocal with resonance.
Notch	Cuts a band out. Resonance narrows it. Slope is ignored.
Peak	Boosts a band. Resonance is the boost, up to five times.
Ladder	A four-pole transistor ladder: warm, thick, and it squelches and self-oscillates near full resonance.
Comb +	A short tuned delay with positive feedback: rings at every harmonic of the cutoff, like a plucked string or a tube.
Comb –	Negative feedback: rings at the odd harmonics, hollower, an octave lower.

Routing

Serial Filter 1 into Filter 2. Filter 2 starts as an open ladder, so it changes nothing until you turn it down, and costs nothing while it stays open.

Parallel Each filter hears what goes into it, and **BALANCE** turns one side down.

Each oscillator's **FILTER**, and the noise's, picks where it goes: **F1**, **F2** or both (**1+2**, the default). In Serial, F1 and 1+2 go through both filters and F2 skips Filter 1. In Parallel, a source in one filter plays at full level with BALANCE at centre, and turning BALANCE the other way turns it down. A source in both crossfades: half into each at centre, all into one at either end.

> **TWO SOUNDS IN ONE**
> Set Osc 1 to **F1** and Osc 2 and 3 to **F2**, in Parallel, to filter a body and a bright layer apart: a low pass on one, a comb or high pass on the other.

Drive

The drive sits before the filters. **Soft** is a tanh curve, round and warm. **Hard** clips flat, buzzier. **Tube** is asymmetric, so it adds even harmonics. Its gain is compensated, so turning it up thickens and compresses the sound more than it makes it louder.

CHAPTER 10

Envelopes and LFOs

Three envelopes and four LFOs per voice. Two of the LFOs move the creatures as well as the sound.

BOTTOM LEFT

ENVELOPES



Each envelope draws its shape: drag the points to set **attack**, **decay**, **sustain** and **release**. A point moves with the mouse from where you grab it; hold **Shift** for fine moves. A dot rides the curve for every sounding note. The chip above each one is its source for MAP and lights as it moves.

ENV 1 AMP is the amp: it shapes every note's level, and it always releases when you let go. **ENV 2** and **ENV 3** are free to route; a new part starts with Env 2 opening Filter 1.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
Attack	`0 ms – 20 s`	`5 ms`	ROUTE
Decay	`0 ms – 20 s`	`400 ms (2, 3: 300 ms)`	ROUTE
Sustain	`0 – 100%`	`70% (2, 3: 30%)`	ROUTE
Release	`0 ms – 20 s`	`300 ms`	ROUTE
Time Range	`Short, Normal, Long`	`Normal`	How much time the drawing spreads across its width: up to 1 s, 5 s or 20 s for each of attack, decay and release. Pick **Short** for plucks and drums, **Long** for pads and swells. It only changes the view: switching it never changes a time, and a time past the end waits there until you drag it back in.
Loop	`Off, Held, Free`	`Off`	**Held** repeats attack and decay while the note is held. **Free** keeps repeating after you let go too, until the note ends.

VELOCITY on the VOICE tab under the sky (0 to 100%, 100% by default) is how much velocity sets ENV 1's level. At 0 every note plays at full level.

Host parameters per envelope: **Delay** and **Hold** (0 ms to 20 s) ROUTE, an **Attack**, **Decay** and **Release Curve** each (−1 to +1), and **Velocity**, how much velocity scales the whole envelope.

UNDER THE SKY

MOTION



The MOTION tab holds the two LFOs that move the flock. They only move it: they aren't sources for routes. For a sound that moves with them, use LFO 1 or 2 at the same rate.

WINGBEAT One per creature (5 Hz sine by default). It bobs each creature across its path, by **PUSH**, and beats its wings on screen.

WIND One gust for the whole sky (a slow 0.2 Hz Smooth, Global by default). It pushes every creature of the part the same way, by **PUSH**, from a direction that turns slowly on its own.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
Shape	`Sine, Triangle, Saw, Square, S&H, Smooth`	`Sine (Wind: Smooth)`	S&H holds a random value per cycle; Smooth glides between random values.
Trigger	`Retrigger, Free, Global`	`Retrigger (Wind: Global)`	See below.
FREE / SYNC	`Rate or tempo`	`Free`	Sync follows the host's tempo at **DIVISION**; Free and Global also lock to its bar.
RATE	`0.02 – 50 Hz`	`5 Hz (Wind: 0.2 Hz)`	ROUTE
PUSH	`0 – 100%`	`20% (Wind: 30%)`	How hard it moves the creatures. ROUTE
PHASE	`0 – 360°`	`0°`	Where a retriggered cycle starts, or how far a synced one sits off the beat. ROUTE
DELAY	`0 ms – 20 s`	`0`	Fades the LFO in after each note starts. ROUTE
DIVISION	`1/32 – 4/1`	`1/4`	With triplets and dotted values.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
POLARITY	`Bipolar, Unipolar`	`Bipolar`	Bipolar swings both ways around the knob; unipolar only up from it.

ENVELOPES

LFOS



LFO 1 (4 Hz sine) and **LFO 2** (0.5 Hz triangle) are plain LFOs that only modulate. The **LFOS** tab draws each one, with a dot running along it at its rate. They have the same controls as the MOTION LFOs, without Push.

TRIGGER

Retrigger	Each note's LFO starts at Phase. Notes are in step with how you play them.
Free	Each note's LFO picks up where the part's running LFO is, so notes are in step with each other.
Global	One LFO for the whole part. Every note hears the same value; a route can't change its rate per note.

ON THE GRID

With **SYNC** on, a **Free** or **Global** LFO is locked to your DAW's bar, not just its tempo: a 1/16 gate lands on the sixteenths wherever you start playback, and a looped bar plays the same S&H values every time round. **PHASE** shifts it against the beat. **Retrigger** starts it with each note instead, which suits notes played off the grid.

VIBRATO

Route LFO 1 to the part's **Fine** at a few percent for vibrato, and set its **DELAY** so it arrives once the note has settled. Map Press onto that route's amount and the vibrato follows your finger.

CHAPTER 11

The Flock

How each species flies, and what flying near another creature does to its sound.

UNDER THE SKY

FLOCK



CONTROL	RANGE	DEFAULT	WHAT IT DOES
Behaviour	`Shoal, Scatter, Hunt, Flee`	`Shoal (B: Hunt)`	See below.
COHESION	`0 – 100%`	`50%`	Pull toward the middle of the neighbours. ROUTE
ALIGN	`0 – 100%`	`50%`	Turn to fly the way the neighbours fly. ROUTE
SEPARATE	`0 – 100%`	`40%`	Keep clear of the nearest. ROUTE
SPEED	`0 – 100%`	`40% (B: 55%)`	Top speed. Press raises it further, up to 80% more. ROUTE
WANDER	`0 – 100%`	`20%`	Each creature's restlessness, and how far it roams around its lure. ROUTE
LURE	`0 – 100%`	`60%`	How hard your note pulls its creature. Low lets the flock take over. ROUTE
RANGE	`0 – 100%`	`35%`	How near counts as a neighbour. ROUTE
TAIL	`0 – 100%`	`50%`	How long the ink lasts, about 0.3 to 3 seconds. ROUTE
SPREAD	`0 – 100%`	`50%`	How far the ink blooms as it ages. ROUTE
SIZE	`0 – 100%`	`50%`	How big the sky is. A bigger sky makes creatures relatively smaller and slower, so flocks spread out. Shared.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
WALLS	`Soft, Wrap, Open`	`Soft`	Soft turns creatures back near the edge; Wrap carries them round to the other side; Open lets them fly well past it first. Shared.
LURE X, LURE Y	`Pitch, Velocity, Press, Slide, Release, Random, Centre`	`Pitch, Slide`	What places a note's lure along each axis.

Behaviours

Shoal Flies with its own kind: cohesion, alignment and separation as set.

Scatter Keeps apart: cohesion pushes creatures away from their neighbours' middle instead of toward it.

Hunt Chases the nearest creature of the other species. A hunter that gets close enough catches it, which turns on its Bite. Everything shies away from hunters a little.

Flee Runs hard from the other species, whatever it is doing.

The lure

A held note pulls its creature toward a point that roams slowly around the lure, on a leash: loose near the lure, so the flock and the wander shape the path there, and firm when the creature strays. Each creature roams on its own two slow orbits, so no two trace the same path. **LURE X** and **LURE Y** choose what places the lure: by default low notes sit left and high notes right (Pitch, over four octaves from C2, and bend moves it), and more Slide sits higher. Try Lure Y on *Press* to make pressing harder send creatures upward, or Lure X on *Random* to scatter a chord across the sky.

Between creatures

Three knobs on the **FM** tab connect the flock to the sound directly. All start at 0, so the flock changes nothing until you ask it to.

FLOCK FM Each creature listens to up to three neighbours of its own species, weighted by how near they are, and their Osc 1 frequency-modulates its Osc 2. A tight flock rings; a loose one is clean. When a creature gains or loses a neighbour, the change fades in over about 10 ms, so it never clicks. For it to be heard, Osc 2 must be in the sound: give it level, or route it into Osc 1 on the FM grid.

TUNE PULL For each neighbour heading within 30° of the same way, the creature is pulled toward the nearest just interval from that neighbour's pitch (unison, minor third, major third, fourth, fifth, major sixth, octave), up to ±30 cents at 100%, over about 150 ms. Chords played on a flock that flies together settle into just intonation; break formation and they drift back.

BITE While one of this part's hunters holds a creature of the other species, the hunter's Osc 1 ring-modulates it, by Bite times how caught it is. Set Bite on the hunting part. Because a catch lets go over half a second, each bite rings out as the prey escapes.

Ink

Every creature leaves ink behind it in the sky, more the brighter it is. The ink fades with **TAIL** and blooms with **SPREAD**. It is only drawn: it doesn't change the sound.

CHAPTER 12

Touch

Five dimensions of every note, each shaped and smoothed, each a source.

UNDER THE SKY

TOUCH



TOUCH	WHERE IT COMES FROM
VELOCITY	How hard the note was struck: its velocity.
PRESS	Pressure: MPE channel pressure or the CLAP pressure expression. It also makes the creature faster and brighter.
GLIDE	The finger's bend, as −100% to +100% of the bend range.
SLIDE	Up and down the key: MPE CC 74 or the CLAP brightness expression.
RELEASE	How fast the note was released: its release velocity, 0 while the note is held. Route it to the amp envelope's release, say, so a quick lift cuts the note short.

Each has a **CURVE** (−1 to +1): positive bows it up, so it responds more early in the travel, and negative bows it down. And a **SMOOTH** time (8 ms by default, 2 ms for Glide so slides track the finger), which irons out a controller's steps.

Under them, the rest of how you play. The whole tab is shared by both parts.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
ROUND	`0 – 100%`	`0%`	Glide Rounding: once a finger stops sliding, pull its pitch toward the nearest semitone, by this much. While the finger moves, the pitch follows it exactly.
ROUND TIME	`0 ms – 20 s`	`150 ms`	How long that pull takes.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
MGR BEND	`1 – 96 st`	`2 st`	The bend range of the MPE manager channel (channel 1), which bends every note.

The touch chips

The five chips in the header light as you play, showing the newest note of the part you are editing. In MAP mode they are sources like any other: touch **PRESS** and drag a knob to make pressure move it.

> **NO MPE CONTROLLER?**
> Velocity comes from any keyboard. A plain keyboard's channel pressure is Press for every note, and its pitch wheel bends every note by MGR BEND. In the sky, a click sets Slide by how high you click and dragging bends.

CHAPTER 13

Modulation: MAP and Routes

There is no matrix page. Touch a source, drag a knob.

Mapping

01 **Touch a source**

Click any source chip: the touch chips in the header, the envelope and LFO chips on their panels, or the flock's under the sky. That turns MAP on: the button lights pink, the source chips pulse, the one you touched lights, and the sky tells you to drag a knob. (Pressing **MAP** first works too.)

02 **Drag a knob**

Drag any knob up or down to set how much the source moves it, from −100% to +100%. A ring in the source's colour shows how far it reaches. Drag more knobs to send the same source to them too, or touch another source.

03 **Press MAP again**

To finish. Knobs go back to setting their own values, and the rings go away.



Fig. 13.1 MAP with a source armed: the knobs it moves show rings in its colour.

IN MAP MODE, WITH A SOURCE ARMED	ON A KNOB
Drag	Set that source's amount on this knob. A route dragged back to 0 is removed.
Double-click	Flip the route's sign.
Right-click	Remove the route. With no source armed, removes every route to that knob.

Only the armed source's rings show; touch another source to see its routes. Nearly every knob on a part's panels can take routes: the flock's knobs, the FM grid, the envelopes and LFOs, the filters and the oscillators. The exceptions are Transpose, the part's Bend range, Velocity, and the shared ones: Size, Width, Master and the TOUCH tab.

Sources

SOURCE	VALUE
Env 1, 2, 3	0 to 100%, each note's own.
LFO 1, 2	−100% to +100% (or 0 to 100% when Unipolar). Per note unless Global.
Velocity, Press, Slide	0 to 100%, each note's own.
Release	The note's release velocity, 0 to 100%: 0 while it is held.
Glide	−100% to +100%: the note's bend.
Key	The note: −100% four octaves below middle C, 0 at middle C, +100% four octaves above.
Speed, Nearest, Heading	The flock's own, 0 to 100% (Chapter 6).
Turn	The flock's turning, −100% (left) to +100% (right) (Chapter 6).
Random	A new value per note, −100% to +100%.
Mod Wheel	CC 1 on channel 1.
Macro 1 to 4	Four host parameters, for automation or a controller.

Key, Mod Wheel and the macros have no chip on the panel; choose them in the route list. Random's chip is under the sky with the flock's.

The route list

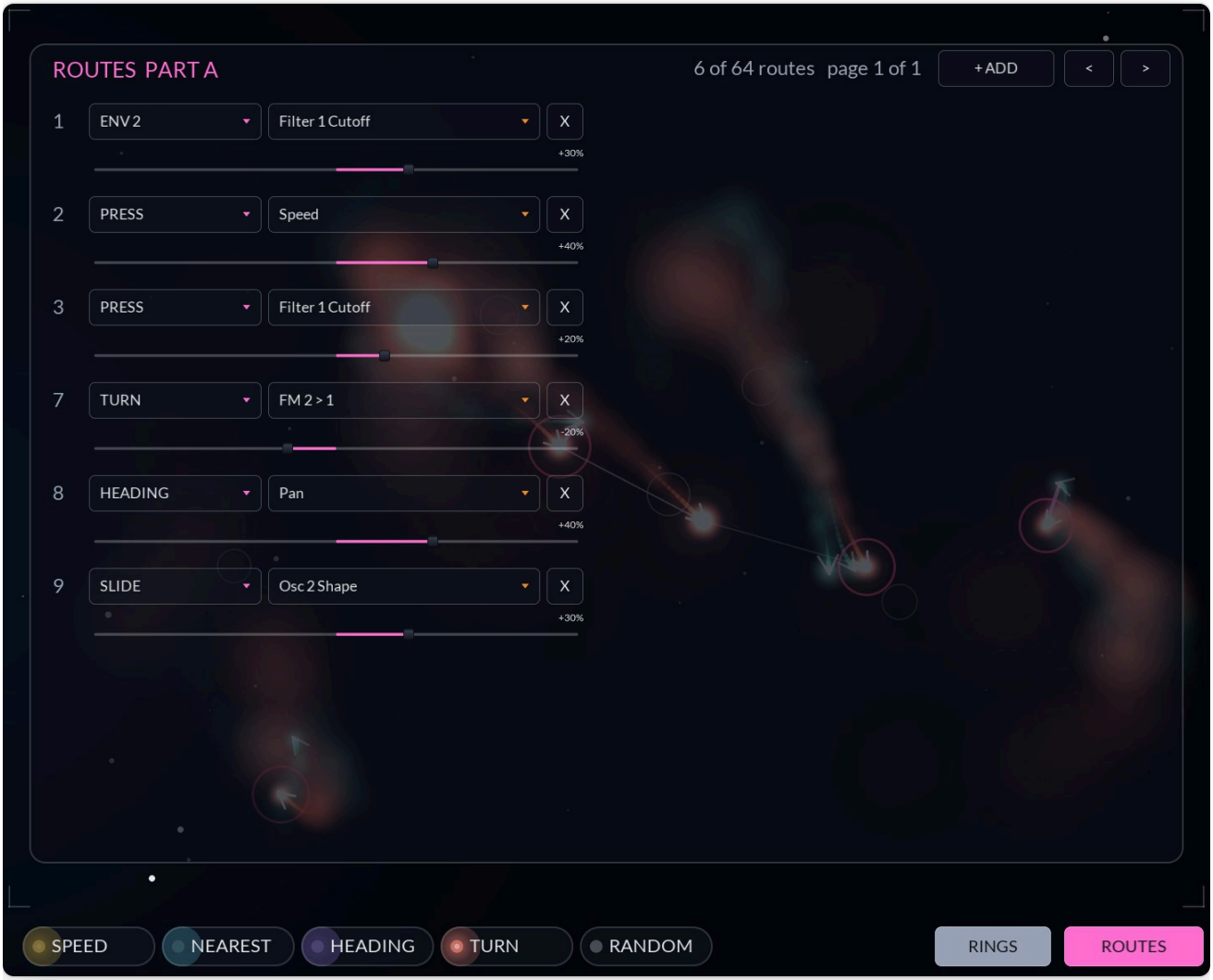


Fig. 13.2 ROUTES: every route of the part in use, with its source, destination and amount.

ROUTES under the sky opens, over the sky, the list of the part's routes in use, sixteen to a page. Each row has the source, the destination and the amount; change any of them here. **+ ADD** starts a new route (Env 2 onto Filter 1 Cutoff, at 0) to edit, and **X** removes one. The arrows turn the pages. Each part has 64 routes; the list shows how many are used.

The list reaches what MAP can't: sources without a chip, and destinations with no knob on the panel, such as an oscillator's Phase, an envelope's Delay and Hold, and every route's own amount.

Routes onto routes

A route's amount is itself a destination. Route Press onto the amount of a route from LFO 1 to Fine, and the vibrato deepens as you press. Routes onto amounts are worked out first, so they always take effect on the same tick.

How far a route moves a knob

A route moves a parameter within its own range: +100% from a source at full moves a knob from its lowest setting to its highest, and +10% moves it a tenth of the way. Routes add together, and the result stops at the knob's ends. Ratio moves by musical intervals, so a small route onto Ratio detunes evenly at every setting.

CHAPTER 14

Factory Presets

Ninety-two sounds in fourteen categories: the instrument families first, then patches that move on their own, use both parts, or let the flock play.

The preset bar works like JVerb's: the arrows step through the presets, scrolling over the bar does too, and clicking the name opens the menu by category. **SAVE** saves the current sound under a name of your own. A factory preset sets every parameter it doesn't mention to its default, so nothing from the last sound (a route, say) carries over. Within each category, the presets run from a plain patch up to ones with more going on, and most answer the mod wheel and Press. Every preset is level-matched, so stepping through the bank doesn't jump.

7 PRESETS

Bass

Mono, speaking at once; the wheel brightens or growls.

Sub Bass	A sine with a touch of feedback and a quiet octave; the wheel adds feedback.
Analog Bass	Saw and a square an octave down through a ladder; Env 2 plucks it, harder when struck harder.
FM Bass	A DX bass: 1:1 FM with feedback on the modulator, its depth dying away; Velocity bites.
Funk Bass	Saw and pulse, a short velocity-scaled filter pluck.
Acid Bass	Saw into a resonant, driven ladder with glide between held notes; Velocity accents, Slide opens it.
Reese Bass	Two saws 22 cents apart over a sub, LFO 2 drifting the detune; drive.
Wobble Bass	Saw and square through a tube-driven ladder, an eighth-note wobble at the host's tempo; the wheel deepens it.

7 PRESETS

Leads

Mono with glide; the wheel deepens a vibrato, Press opens the sound.

Saw Lead	Two detuned saws, vibrato after 350 ms.
Square Lead	Square over a square an octave down, LFO 2 on the width.
Sync Lead	Osc 2 synced to a silent Osc 1; Env 2, the wheel and Press sweep its pitch.
Soft Flute	Sine and a soft triangle with a breath of noise at the start and under Press.
FM Lead	1:1 FM with feedback; Press and Velocity open it.
Pressure Lead	Press opens 2:1 FM and brings in vibrato; Slide adds feedback grit.

Overdrive Lead	Saws and a sub driven hard into a resonant ladder; Press drives harder.

7 PRESETS

Keys

Played like a keyboard: pianos dying away, organs held.

Tine Piano	A DX electric piano: a 1:1 pair and a ×14 tine, both harder when struck harder; high notes die sooner.
Suitcase EP	The tine piano through a stereo tremolo the wheel widens; Press pushes the bark.
Reed Piano	A half-sine modulator that barks when struck hard, through a little drive; the wheel adds tremolo.
Clav	Narrow pulse through a resonant filter, Velocity on the cutoff.
Harpsichord	Saw and a narrow pulse an octave up, high passed, with a short release.
Tonewheel Organ	Drawbars 16′, 8′ and 5 1/3′, key click, and a rotary speaker the wheel speeds up.
Church Organ	Sines at 16′, 8′ and 4′ with a slow speak and a long release.

6 PRESETS

Mallets and Bells

Struck, ringing out.

Marimba	×4 and ×10 partials; Env 3 and Velocity bring in the ×10.
Vibraphone	×4 FM with a tremolo the wheel deepens.
Celesta	A sine with a ×7 modulator and a long decay.
Tubular Bells	A ×3.5 modulator over a long ring, a second carrier an octave up.
Crystal Bells	Inharmonic ×1.41 and ×7 with a slow shimmer on the ×1.41; the wheel deepens it.
Steel Drum	×3 and ×2 modulators, slightly detuned, opened by Env 3 and Velocity.

8 PRESETS

Plucks

Struck, gone in a second or two.

Synth Pluck	Saw and detuned pulse through a resonant filter; Slide on the resonance, the wheel lengthens it.
Bright Pluck	Two saws panned apart and a square an octave up; the wheel lets it ring longer.
Nylon Guitar	Saw through a key-tracked low pass, Env 2 on the cutoff.
Harp	1:1 FM with a long ring.
Koto	×3 modulator with feedback, Env 3 and Velocity on the FM.
Kalimba	×3 FM blip with an inharmonic ×5.95 partial.

Pizzicato	Two saws, a short, key-tracked filter pluck.
Bell Pluck	×3.5 and ×7; Press on the upper FM, LFO 1 on the detune.

4 PRESETS

Brass

The filter opens a beat after the note speaks.

Synth Brass	Two detuned saws, Env 2 on the filter (deeper when struck harder), a pitch scoop, vibrato on the wheel.
Soft Horns	Saw and triangle with a slow speak.
East Coast Brass	Feedback-driven 1:1 FM with a slow Env 3; Press opens it.
Brass Stabs	Three saws, short and bright, driven; Velocity sets the bite.

4 PRESETS

Strings

Bowed: a slow speak and a vibrato that waits.

String Machine	Three saws with slow detune LFOs; Press and the wheel open the filter.
Ensemble Strings	Saws drifting against each other on LFO 2; a harder note bows in faster.
Slow Strings	The ensemble over a second and more, opening as it swells.
Solo Cello	Mono saw and pulse with glide; Press brings in vibrato and opens the filter.

8 PRESETS

Pads

Rising and held; the wheel and Press open most of them.

Sine Pad	Two detuned sines panned apart, no routes.
Warm Analog	Saw pair through a low pass, slow Env 2.
PWM Pad	Two squares whose widths drift on unrelated LFOs.
Glass Pad	×3 and ×0.5, LFO 1 on the FM.
Choir Pad	Saws through two band passes in parallel for an "ah"; a slow LFO moves the vowel, the wheel darkens it.
Bowed Glass	2:1 FM that blooms with a slow Env 3.
Dawn Chorus	Vibrato that arrives once the note has bloomed; Press on the FM.
Deep Current	A sub through a resonant, tube-driven ladder; Slide drives it harder.

4 PRESETS

Rhythmic

Moving in time with the host's tempo.

Trance Gate	Saws gated in sixteenths; the wheel softens the gate.
Sixteenth Pulse	A falling saw LFO snaps a ladder open each sixteenth, so a held note plays a line.
Sample and Hold	A new random cutoff each sixteenth through a resonant ladder.
Dotted Chords	Chords chopped on dotted eighths, the filter breathing on the half note.

4 PRESETS

Atmospheres

Seconds to arrive.

Wind Choir	Saws and noise through a band pass in a strong wind; LFO 2 on the cutoff.
Deep Space	Pink noise through a comb tuned to the note, slow random LFOs on its filters.
Aurora	A slow FM shoal with Tune Pull; Heading on the FM, Slide on the cutoff.
Storm Front	Saw and noise through a comb in a strong wind; LFO 2, Turn and Speed on the filters.

7 PRESETS

Drums and FX

Every key the same drum, or a sound effect.

Kick	A sine whose pitch drops two octaves in 50 ms.
Snare	Noise and a body tone with a snap.
Hi-Hat	White noise high passed; the wheel opens it.
Wood Block	A sine with a ×2.76 modulator, 70 ms.
Log Drum	Self-feedback and a pitch drop that goes further the harder you strike.
Laser Zap	FM with feedback, its pitch diving.
Riser	Hold a note: saws and noise rise an octave and open over four seconds.

6 PRESETS

Motion

Patches that keep moving on their own.

Shifting Glass	Three-operator FM with a looping Env 3 and two unsynced LFOs; the wheel deepens all of it.
Random Garden	The Random wave: each note draws its own waves, LFO speed and pan.
Pulsar	A looping Env 1: every held note pulses on its own clock, faster higher up and with Press.
Clockwork	A sixteenth-note sample-and-hold on the FM and a dotted-eighth square on the filter.

MPE Voices	Two band passes for a vowel: Press brightens and adds vibrato, Slide moves the vowel, Release sets the tail.
Tide Machine	Layer: a saw drone breathing on slow LFOs, and FM bells an octave up looping on their own envelope.

7 PRESETS

Layers and Splits

Both parts at once.

Piano and Strings	Layer: the tine piano, and strings under it the wheel brings up.
Bell Pad	Layer: an FM bell for the strike over a warm pad that swells in behind it.
Bass and Keys Split	Split at C3: an analog bass below, the reed piano above.
Bass and Lead Split	Split at C3: an acid bass below, a saw lead above.
Velocity Stab	Velocity: play softly for a pad, dig in for a brass stab.
Breath to Brass	Expression: Press crossfades each note from a breathy flute to bright brass.
Octave Shimmer	Layer: a warm pad and an octave-up ring of sine bells with Neighbour FM and Tune Pull.

13 PRESETS

Flock

The creatures play the sound: from one species up to both, hunting each other.

Fireflies	Half Sine blips; a fast, scattered flock with short ink, spread wide.
Metal Rain	Inharmonic ×3.73 and ×7.11, scattered; each note detuned at random.
Starling Bells	Neighbour FM, Tune Pull and a cohesive flock; Nearest deepens the FM.
Dripping Ink	Long, wide ink; Turn, Speed and Heading play the filter and FM.
Ink Ribbons	Long ink and a loose lure; Turn plays cutoff and FM.
Murmuration Pad	Saws, an aligned flock spread wide; Heading pans each voice.
Pressure Front	Press opens 2:1 FM, with Neighbour FM.
Hunter Clicks	Layer: A flees, B hunts an octave up with Bite and Neighbour FM.
Mallet Flock	Layer: vibes and an upper shoal with Neighbour FM and Tune Pull.
Shark Bass	Split at C3: below, a hunting saw-and-pulse bass with Bite; Speed opens its filter.
Predator and Prey	Layer: A flees, B hunts an octave down; A's filter opens as its own kind closes in.
Murmuration Riot	Layer: a fast FM shoal opened by Press, Speed, Nearest and Turn, and driven ladder hunters that bite.
Murmuration Storm	Layer: a saw shoal with Neighbour FM and Tune Pull, and FM hunters.

INIT

Init is every parameter at its default: a saw through a low pass on part A, with the three starting routes (Env 2 and Press on the cutoff, Press on the flock's speed).

Your presets

User presets are small text files ending `.jspreset`, saved in:

macOS	`~/Library/Application Support/JSynth/Presets`
Windows	`%APPDATA%\JSynth\Presets`
Linux	`~/.config/JSynth/Presets (or $XDG_CONFIG_HOME/JSynth/Presets)`

Copy them between machines freely. A preset from a newer JSynth loads in an older one with any parameters it doesn't know skipped.

CHAPTER 15

Sound Design Recipes

Starting points that show off what the flock can do. Each starts from Init unless it says otherwise.

DX electric piano

Two sines, a 1:1 pair for the body and a ×14 for the bark.

```
Osc 1 Sine · Osc 2 Sine Ratio ×1 Level 0
Osc 3 Sine Ratio ×14 Level 0
Env 1 A 1 ms D 5 s S 0 R 400 ms
Env 2 → FM 2 > 1 +30% · Env 3 (D 60 ms) → FM 3
> 1 +35%
Velocity → FM 2 > 1 +20% · Filter 1 open
```

Or load *Tine Piano*.

A flock that rings

Neighbours modulate each other, so a held chord shimmers as it gathers and breaks.

```
Osc 2 Sine Ratio ×3.5 · FM 2 > 1 40%
FLOCK FM 70% · TUNE PULL 40% · COHESION 70%
Nearest → FM 2 > 1 +30%
X PAN 60%
```

Or load *Starling Bells*.

Predator and prey

Two species; every catch rings, and the prey's filter opens as its own kind closes in.

```
Play mode LAYER
A: Flee · Nearest → F1 Cutoff +60%
B: Hunt · Speed 70% · Pulse · BITE 90% · Transp
-12
```

Or load *Predator and Prey*.

Turning pad

The sound leans as each creature banks.

```
Osc 1 Alt Sine · FB 1 25%
TAIL 100% · SPREAD 90% · LURE 35%
Turn → F1 Cutoff +50% · Turn → FM 2 > 1 +40%
```

Or load *Ink Ribbons*.

Wind through a choir

A strong gust pushes the flock while LFO 2 sweeps the filter.

```
Saws, Noise 35% · F1 Band Pass, Res 40%
Wind Push 90% · Wind Rate 0.1 Hz
LFO 2 → F1 Cutoff +40% · Slide → F1 Res +30%
```

Or load *Wind Choir*.

Pressure vibrato

Vibrato only where you press, arriving after the note settles.

```
LFO 1 5.5 Hz, DELAY 350 ms
Route LFO 1 → Fine at 0
Press → that route's amount +6%
```

As in *Pressure Lead*.

A stereo murmuration

The flock spreads a pad across the speakers as it turns.

```
Two saws, detuned · ALIGN 80% · SPEED 25%
X PAN 80% · DRIFT 15%
Heading → Pan +60%
```

Or load *Murmuration Pad*.

Struck drum

A pitch drop that goes further the harder you hit.

```
Osc 1 Sine · FB 1 15% · Env 1 D 300 ms S 0
Env 3 (D 45 ms) → Osc 1 Ratio +10%
Velocity → that route's amount +10%
```

Or load *Log Drum*.

APPENDIX A

Parameter Reference

Every parameter JSynth exposes to the host. Part parameters exist once for each part, named "A …" and "B …" in the host.

GLOBAL

PARAMETER	RANGE	DEFAULT
Play Mode	Single, Layer, Split, MPE Zones, Velocity, Expression	Single
Split Key	C-1 – G9	C4
Velocity Split	1 – 127	80
Crossfade Source	Slide, Press	Slide
Quality	1x, 2x, 4x	2x
Master Level	-60 – +6 dB	-6.0 dB
Output Width	0 – 200%	100%
Manager Bend Range	1 – 96 st	2 st
Glide Rounding	0 – 100%	0%
Rounding Time	0 ms – 20 s	150 ms
Macro 1 – 4	0 – 100%	0%
Velocity … Release Curve	-1 – +1	+0.00
Velocity … Release Smoothing	0 ms – 20 s	8.0 ms (Glide 2.0 ms)
Size	0 – 100%	50%
Walls	Soft, Wrap, Open	Soft

VOICE (PER PART)

PARAMETER	RANGE	DEFAULT	
Level	-60 – +6 dB	0.0 dB	ROUTE
Pan	-100% – +100%	+0%	ROUTE
X Pan	0 – 100%	0%	ROUTE
Transpose	-48 – +48 st	+0 st (B: -12 st)	
Fine	-100 – +100 ct	+0 ct	ROUTE
Key Low, Key High	C-1 – G9	C-1, G9	

PARAMETER	RANGE	DEFAULT	
Polyphony	1 – 16	8	
Voice Mode	Poly, Mono, Legato	Poly	
Glide	0 ms – 20 s	0.0 ms	ROUTE
Glide Mode	Always, Legato	Always	
MPE Bend Range	1 – 96 st	48 st	
Drift	0 – 100%	0%	ROUTE
Velocity	0 – 100%	100%	
Output	Main, Own	Main	

OSCILLATORS (PER PART: OSC 1, 2, 3)

PARAMETER	RANGE	DEFAULT	
Wave	Sine, Triangle, Saw, Square, Pulse, Half Sine, Abs Sine, Quarter, Alt Sine, Alt Half, Camel, Alt Camel, White, Pink	Saw (Osc 2, 3: Sine)	
Shape	0 – 100%	50%	ROUTE
Pitch Mode	Note, Ratio, Fixed	Note (Osc 2, 3: Ratio)	
Coarse	−48 – +48 st	+0 st	ROUTE
Fine	−100 – +100 ct	+0 ct	ROUTE
Ratio	×0.25 – ×16	×1 (Osc 3: ×3)	ROUTE
Fixed	1 Hz – 10 kHz	440 Hz	ROUTE
Sync	Off, On	Off	
Retrigger	Off, On	On	
Phase	0 – 360°	0°	ROUTE
Bend	0 – 100%	100%	
Key Track	Off, On	On	
Filter	1+2, F1, F2	1+2	
Level (mixer)	0 – 100%	100% (Osc 2, 3: 0%)	ROUTE
Pan (mixer)	−100% – +100%	+0%	ROUTE

FM, NOISE AND DRIVE (PER PART)

PARAMETER	RANGE	DEFAULT	
FM Feedback 1 – 3, FM 1 > 2 … 3 > 2	0 – 100%	0%	ROUTE
FM Type	Phase, Exponential	Phase	
Noise Level	0 – 100%	0%	ROUTE

PARAMETER	RANGE	DEFAULT	
Noise Color	-100% – +100%	+0%	ROUTE
Noise Filter	1+2, F1, F2	1+2	
Drive Type	Soft, Hard, Tube	Soft	
Drive	0 – 100%	0%	ROUTE

FILTERS (PER PART: FILTER 1, 2)

PARAMETER	RANGE	DEFAULT	
Filter Routing	Serial, Parallel	Serial	
Filter Balance	-100% – +100%	+0%	ROUTE
Type	Low Pass, High Pass, Band Pass, Notch, Peak, Ladder, Comb +, Comb –	Low Pass (F2: Ladder)	
Slope	12, 24, 36, 48 dB	24 dB	
Cutoff	20 Hz – 20 kHz	1.26 kHz (F2: 20.00 kHz)	ROUTE
Resonance	0 – 100%	20% (F2: 0%)	ROUTE
Drive	0 – 100%	0%	ROUTE
Key Track	0 – 100%	0%	ROUTE

ENVELOPES (PER PART: ENV 1, 2, 3)

PARAMETER	RANGE	DEFAULT	
Delay	0 ms – 20 s	0.0 ms	ROUTE
Attack	0 ms – 20 s	5.0 ms	ROUTE
Hold	0 ms – 20 s	0.0 ms	ROUTE
Decay	0 ms – 20 s	400 ms (Env 2, 3: 300 ms)	ROUTE
Sustain	0 – 100%	70% (Env 2, 3: 30%)	ROUTE
Release	0 ms – 20 s	300 ms	ROUTE
Attack, Decay, Release Curve	-1 – +1	+0.00	
Loop	Off, Held, Free	Off	
Velocity	0 – 100%	0%	
Time Range	Short, Normal, Long	Normal	

LFOS (PER PART: WINGBEAT, WIND, LFO 1, LFO 2)

PARAMETER	RANGE	DEFAULT
Shape	Sine, Triangle, Saw, Square, S&H, Smooth	Sine (Wind: Smooth, LFO 2: Triangle)

PARAMETER	RANGE	DEFAULT	
Rate	0.02 – 50 Hz	5.00, 0.20, 4.00, 0.50 Hz	ROUTE
Sync	Off, On	Off	
Division	1/32 – 4/1	1/4	
Trigger	Retrigger, Free, Global	Retrigger (Wind: Global)	
Phase	0 – 360°	0°	ROUTE
Delay	0 ms – 20 s	0.0 ms	ROUTE
Polarity	Bipolar, Unipolar	Bipolar	
Push (Wingbeat, Wind)	0 – 100%	20%, 30%	ROUTE

FLOCK (PER PART)

PARAMETER	RANGE	DEFAULT	
Behaviour	Shoal, Scatter, Hunt, Flee	Shoal (B: Hunt)	
Cohesion	0 – 100%	50%	ROUTE
Alignment	0 – 100%	50%	ROUTE
Separation	0 – 100%	40%	ROUTE
Speed	0 – 100%	40% (B: 55%)	ROUTE
Wander	0 – 100%	20%	ROUTE
Lure	0 – 100%	60%	ROUTE
Range	0 – 100%	35%	ROUTE
Neighbour FM	0 – 100%	0%	ROUTE
Tune Pull	0 – 100%	0%	ROUTE
Bite	0 – 100%	0%	ROUTE
Tail Length	0 – 100%	50%	ROUTE
Tail Spread	0 – 100%	50%	ROUTE
Lure X, Lure Y	Pitch, Velocity, Press, Slide, Release, Random, Centre	Pitch, Slide	

ROUTES (PER PART: ROUTE 1 – 64)

PARAMETER	RANGE	DEFAULT	
Source	Off and 20 sources	Off (1–3: Env 2, Press, Press)	
Destination	159 destinations	Filter 1 Cutoff (2: Speed)	
Amount	−100% – +100%	+0% (1–3: +30, +40, +20%)	ROUTE

APPENDIX B

Specifications

Format	CLAP 1.2 instrument. Features: instrument, synthesizer, stereo.
Ports	One note input (CLAP, MIDI and MIDI MPE; CLAP preferred). Three stereo outputs: Main, Part A, Part B.
Platforms	macOS universal (10.15+ Intel, 11+ Apple Silicon), Windows (64-bit), Linux (64-bit).
Latency	None.
Parts	Two, each a complete patch and a species in the flock.
Voices	Up to 16 per part (40 slots in all, so stolen notes fade out in 3 ms). Mono and Legato with a note stack.
Oscillators	3 per voice, 14 waves, double-precision phase. Saw, square and pulse band-limited by polyBLEP.
FM	3 × 3 grid including feedback; phase or exponential. Plus Neighbour FM from up to three neighbours.
Filters	2 per voice: TPT state variable (12–48 dB), notch, peak, 4-pole ZDF ladder, comb ±. Serial or parallel, each oscillator and the noise picking its filter.
Modulation	3 DAHDSR envelopes with curves and loop, 4 LFOs, 5 touch dimensions, 4 flock sources, key, random, mod wheel, 4 macros. 64 routes per part.
Oversampling	1×, 2× (default) or 4×, decimated by polyphase IIR half-bands.
Control rate	Every 32 samples, ramped per sample. The flock steps at 500 Hz. Renders are identical at any buffer size.
Parameters	All automatable except the route sources, destinations and Quality. Continuous ones accept CLAP modulation; route destinations accept it per note (note id, key, channel or port).
State	Saved with the host project, including the preset name. States from older and newer builds load, with unknown parameters skipped and missing ones at their defaults.
Panel	1648 × 940 at 100%; 75%, 100%, 125%, 150% and 200%.
Threading	The audio thread never allocates, locks or waits on the panel after activation.

Troubleshooting

SYMPTOM	WHAT TO CHECK
JSynth doesn't appear in my host (macOS)	The download is probably quarantined. See the warning in Chapter 2, then rescan plug-ins.
Only part A plays	The play mode is **SINGLE**. Choose **LAYER** or another mode in the header. Check each part's Key Low and Key High too.

SYMPTOM	WHAT TO CHECK
Turning up the flock knobs changes nothing in the sound	By design: the flock only reaches the sound through routes and the three BETWEEN CREATURES knobs. Map a flock source, or raise **FLOCK FM**.
FLOCK FM makes no difference	It modulates Osc 2. Give Osc 2 some **LEVEL**, or route it into Osc 1 with **2 > 1**. Creatures also have to be near each other: raise **RANGE** or **COHESION**, and play more than one note.
BITE does nothing	Bite is set on the hunting part, and needs the hunter's Osc 1 and a prey of the other species: play mode LAYER (or any mode that plays both), one part on **Hunt**.
Notes are out of tune	Check **TUNE PULL** (just intervals are up to 30 cents from equal temperament), **DRIFT**, and a route onto Fine or Ratio. An MPE controller sends bend per note at the part's **BEND** range, which must match the controller's (48 semitones is the MPE standard).
Pitch bend is far too wide on a plain keyboard	A plain keyboard is on channel 1, which bends by **MGR BEND** (2 semitones). If its notes come on other channels, they are taken as MPE member channels and bend by **BEND**.
Notes cut off when I play a lot	The part is at its polyphony and the oldest notes are being taken. Raise the voices count on the VOICE tab (up to 16).
My user presets are missing	Check the preset folder for your OS (Chapter 14). Files must end in `.jspreset`.

Glossary

Creature A note, as it flies in the sky. Each sounding voice has one.

Species A part. Each part's creatures flock with their own kind.

Lure Where a held note pulls its creature.

Boids The classic flocking model (Craig Reynolds, 1986): cohesion, alignment and separation.

MPE MIDI Polyphonic Expression: each note on its own MIDI channel, so each can bend and be pressed on its own.

FM Frequency modulation: one oscillator wobbling another's phase or pitch at audio rate, for bright, bell-like and metallic tones.

Ratio An oscillator's frequency as a multiple of the note's.

Just interval An interval from a simple whole-number ratio, such as 3:2 for a fifth. They beat less than equal-tempered ones.

Route A source, a destination and an amount.

Ring modulation Multiplying two signals, for clangorous sum-and-difference tones.



JSynth

THE FLOCKING SYNTH · V0.9.0

github.com/jeffmhopkins/Jclap